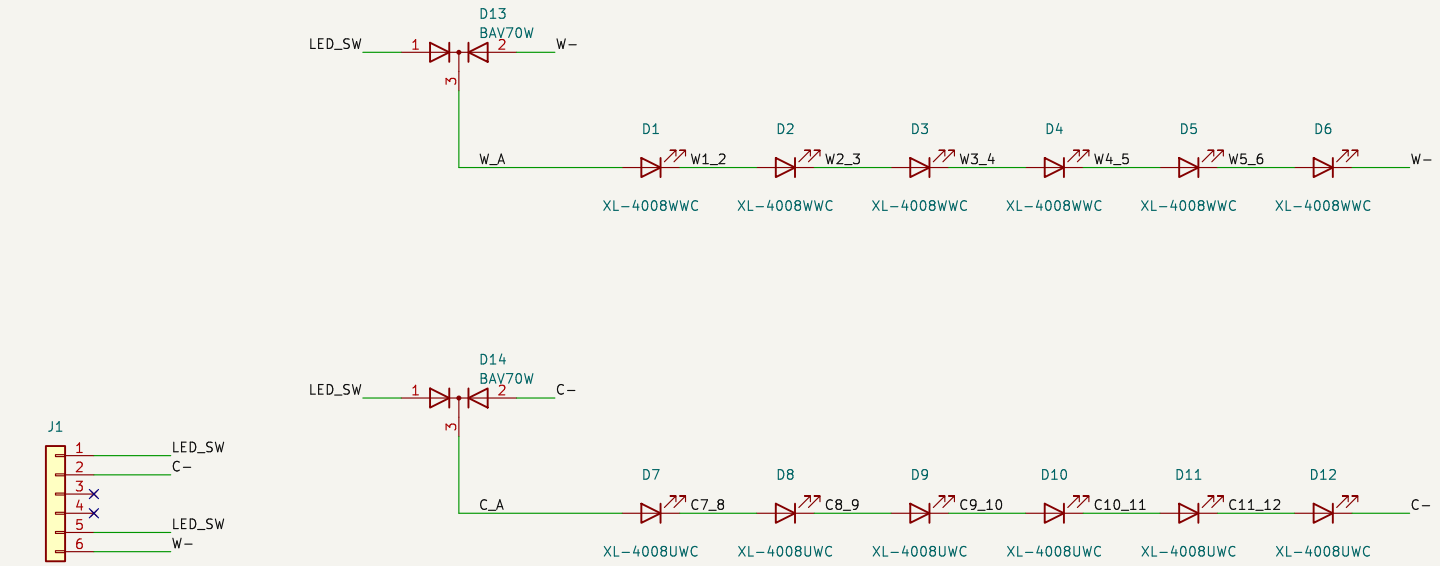


HOW IT CONNECTS
J1 is not a part: it is the flex's own gold-finger tail. It plugs into the main board's frontlight connector
J3 (Hirose FH34SRJ-6S-0.5SH(50)) in place of a panel's bonded frontlight. J3 map (owner-confirmed):
1 C+ = LED_SW, 2 C−, 3 NC, 4 NC, 5 W+ = LED_SW, 6 W−. Pins 1 and 5 are tied here as well (redundant contact).

DRIVER LIMITS (main board U10 TPS923610, R37 = 15 ohm)
Constant current 13.3 mA nominal, 13.9 mA max, one string at a time (Q5 warm / Q6 cool). OVP 24.25 V min;
design ceiling 22 V at −10 C. 6 LEDs per string: about 17.4 V typ (15.0−20.4 V over bins and −10..60 C) + 0.65−0.95 V diode;
LED_SW about 18.4 V typ, 21.7 V worst case. Firmware: 'open' is LED_MONIT >= 2.49 V (peak in the first second).

D13/D14 BAV70W: A1 is the series block, A2 clamps the string if the tail is inserted face-down
(J3 contacts both faces; a flipped tail mirrors pins 1<->6, 2<->5). Flipped: no current, OVP latches.

LEDs: XINGLIGHT XL-4008WWC (3000 K) / XL-4008UWC (6000−6500 K). Warm/cool pairs, top to bottom:
(D7/D1) (D8/D2) (D9/D3) (D10/D4) (D11/D5) (D12/D6); the anode is the top pad of every LED.
On the 3.68in panel the top pair (D7/D1) folds back under the bar on the hinge; it stays in the strings.

Companion to github.com/iandchasse/silkscreen-pcb
(c) 2026 idc LLC
Licensed under CERN-OHL-S-2.0
Warm/cool side-view LED strip on flex for the Silkscreen reader frontlight connector J3
idc LLC

Sheet: /
File: silkscreen-light-strip.kicad_sch

Title: Silkscreen light strip

Size: A4	Date: 2026-09-24	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 1/1